lecture 37

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Ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} + \frac{(z-l)^2}{c^2} = 1 \quad \text{centre } (h,k,l)$$

$$\frac{(x-0)^2}{a^2} + \frac{(y-0)^2}{b^2} + \frac{(z-0)^2}{c^2} = 1 \quad \text{centre } (0,0,0)$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

$$a = b = c$$

$$\frac{x^2}{a^2} + \frac{y^2}{a^2} + \frac{z^2}{a^2} = 1$$

$$x^2 + y^2 + z^2 = a^2 \qquad \times a^2 \text{ on B.S}$$

centre $= C(0,0,0)$, $r = a$

Spheroid

If $a=b$, $a \neq c$, $b \neq c$ spheroid

If $a=b$, $c<a$, $c<b$ oblate

If $a=b$, $c>a$, $c>b$ Polate

Properties

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

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Symmetry

It is symmetric about x-axis, y-axis, z-axis, xy-plane, yz-plane, zx-plane and origin because x, y and z has even power in given equation

(i) If $f(x,y,z) = f(x,-y,-z)$ — x-axis

(ii) If $f(x,y,z) = f(-x,y,-z)$ — y-axis

(iii) If $f(x,y,z) = f(-x,-y,z)$ — z-axis

iv If $f(x,y,z) = f(x,y,-z)$ — xy-plane

v If $f(x,y,z) = f(-x,y,z)$ — yz-plane

vi If $f(x,y,z) = f(x,-y,z)$ — xz-plane

vii If $f(x,y,z) = f(-x,-y,-z)$ — about origin

$f(x,y,z) = 0$, $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

$f(x,-y,-z) = 0$, $\frac{x^2}{a^2} + \frac{(-y)^2}{b^2} + \frac{(-z)^2}{c^2} = 1$

$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

Traces (i) **xy-plane** , $z = 0$

$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is ellips.

(ii) yz-plane , $x = 0$

$\frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ is ellips.

(iii) xz-plane , $y = 0$

$\frac{x^2}{a^2} + \frac{z^2}{c^2} = 1$ is ellips.

Intercept $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

I x-Intercept , $y = 0$, $z = 0$ $(a,0,0)$

$\frac{x^2}{a^2} = 1$, $x^2 = a^2$, $x = \pm a$ $(-a,0,0)$

II y-Intercept $x = 0$, $z = 0$

$\frac{y^2}{b^2} = 1$, $y^2 = b^2$, $y = \pm b$ $(0,b,0)$ $(0,-b,0)$

III z-Intercept $x = 0$, $y = 0$

$\frac{z^2}{c^2} = 1$, $z^2 = c^2$, $z = \pm c$ $(0,0,c)$ $(0,0,-c)$

Example Identify the surface

$100x^2+25y^2+4z^2-400x-150y-8z+529=0$

and find its traces and intercepts.

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Solution

$100x^2-400x+25y^2-150y+4z^2-8z=-529$

$[(10x)^2-2(10x)(20)+(20)^2]+[(5y)^2-2(5y)(15)+(15)^2]+\cdots$

$+[(2z)^2-2(2z)(2)+2^2]=-529+(20)^2+(15)^2+2^2$

$(10x-20)^2+(5y-15)^2+(2z-2)^2=-529+400+225+4$

$(10)^2(x-2)^2+(5)^2(y-3)^2+(2)^2(z-1)^2=100$

$100(x-2)^2+25(y-3)^2+4(z-1)^2=100$

$\frac{100(x-2)^2}{100}+\frac{25(y-3)^2}{100}+\frac{4(z-1)^2}{100}=\frac{100}{100}$

$\frac{(x-2)^2}{1}+\frac{(y-3)^2}{4}+\frac{(z-1)^2}{25}=1$

$\frac{(x-2)^2}{1^2}+\frac{(y-3)^2}{2^2}+\frac{(z-1)^2}{5^2}=1$

is ellipsoid

$\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}+\frac{(z-l)^2}{c^2}=1$

centre $C(h,k,l)=C(2,3,1)$

Traces xy-Plane, $z=0$

$\frac{(x-2)^2}{1^2}+\frac{(y-3)^2}{2^2}+\frac{(0-1)^2}{5^2}=1$

$\frac{(x-2)^2}{1^2}+\frac{(y-3)^2}{2^2}+\frac{1}{25}=1$

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$$\frac{(x-2)^2}{1^2} + \frac{(y-3)^2}{2^2} = 1 - \frac{1}{25}$$

$$\frac{(x-2)^2}{1^2} + \frac{(y-3)^2}{2^2} = \frac{25-1}{25} = \frac{24}{25}$$

$$\frac{25}{24} \times \frac{(x-2)^2}{1} + \frac{25}{24} \times \frac{(y-3)^2}{4} = \frac{24}{25} \times \frac{25}{24}$$

$$\frac{25(x-2)^2}{24} + \frac{25(y-3)^2}{96} = 1 \quad \text{is ellips}$$

yz-Plane $\quad x = 0$

$$\frac{(0-2)^2}{1^2} + \frac{(y-3)^2}{2^2} + \frac{(z-1)^2}{5^2} = 1$$

$$\frac{4}{4} + \frac{(y-3)^2}{2^2} + \frac{(z-1)^2}{5^2} = 1$$

$$\frac{(y-3)^2}{2^2} + \frac{(z-1)^2}{5^2} = 1-4 = -3 \quad \text{No trace}$$

xz-Plane $\quad y = 0$

$$\frac{(x-2)^2}{1^2} + \frac{(0-3)^2}{2^2} + \frac{(z-1)^2}{5^2} = 1$$

$$\frac{(x-2)^2}{1^2} + \frac{9}{4} + \frac{(z-1)^2}{5^2} = 1$$

$$\frac{(x-2)^2}{1^2} + \frac{(z-1)^2}{5^2} = 1 - \frac{9}{4}$$

$$\frac{(x-2)^2}{1^2} + \frac{(z-1)^2}{5^2} = \frac{4-9}{4} = \frac{-5}{4}$$

No trace

Intercept x-Intercept $y=0, z=0$

$$\frac{(x-2)^2}{1^2} + \frac{(0-3)^2}{2^2} + \frac{(0-1)^2}{5^2} = 1$$

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$$(x-2)^2 + \frac{9}{4} + \frac{1}{25} = 1$$

$$(x-2)^2 = 1 - \frac{9}{4} - \frac{1}{25}$$

$$(x-2)^2 = \frac{100-225-4}{100}$$

$$(x-2)^2 = -\frac{129}{100} < 0$$

No x-Intercept

y-Intercept $x=0, z=0$

$$\frac{(0-2)^2}{1} + \frac{(y-3)^2}{4} + \frac{(0-1)^2}{25} = 1$$

$$4 + \frac{(y-3)^2}{4} + \frac{1}{25} = 1$$

$$\frac{(y-3)^2}{4} = 1 - \frac{1}{25} - 4$$

$$= \frac{25-1-100}{25} = -\frac{76}{25} < 0$$

No-y-Intercept

z-Intercept $x=0, y=0$

$$\frac{(0-2)^2}{1} + \frac{(0-3)^2}{4} + \frac{(z-1)^2}{25} = 1$$

$$4 + \frac{9}{4} + \frac{(z-1)^2}{25} = 1$$

$$\frac{(z-1)^2}{25} = 1 - \frac{9}{4} - 4 = \frac{4-9-16}{4}$$

$$= -\frac{21}{4} < 0$$

No z-Intercept

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Example Discuss the symmetry of ellipsoid

$$x^2 + \frac{y^2}{9} + \frac{(z-1)^2}{4} = 1$$

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Solution

(1) **xy - Plane** $z = -z$

$$x^2 + \frac{y^2}{9} + \frac{(-z-1)^2}{4} = 1$$

$$x^2 + \frac{y^2}{9} + \frac{(-1)^2(z+1)^2}{4} = 1$$

$$x^2 + y^2/9 + \frac{(z+1)}{4} = 1$$

$$f(x, y, z) \neq f(x, y, -z)$$

It is not symmetric about xy - plane, x - axis y - axis and origin.

It is symmetric about xz - plane, yz - plane and z - axis because x and y has even power.

yz - Plane $x = -x$

$$(-x)^2 + \frac{y^2}{9} + \frac{(z-1)^2}{4} = 1$$

$$x^2 + \frac{y^2}{9} + \frac{(z-1)^2}{4} = 1$$

xz - Plane , $y = -y$

$$x^2 + \frac{(-y)^2}{9} + \frac{(z-1)^2}{4} = 1$$

$$x^2 + y^2/9 + \frac{(z-1)^2}{4} = 1$$

z - axis $y = -y$, $x = -x$

$$(-x)^2 + (-y)^2/9 + \frac{(z-1)^2}{4} = 1$$

$$x^2 + y^2/9 + \frac{(z-1)^2}{4} = 1$$

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